

PUPIL NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Evidence: *RAMC Stretcher Bearers navigating mud at Passchendaele* • Third Battle of Ypres (1917) **CONTEMPORARY PHOTOGRAPHIC ARCHIVE**

1. FEATURE DRILLS [4M]

Q1(a) & Q1(b): Two 2-mark questions. Identify 1 specific feature + 1 supporting factual detail (5 mins).

2. SOURCE UTILITY [8M]

Q2(a): 3-step evaluation of Content, Provenance (Nature, Origin, Motive), and Contextual Limitations (10 mins).

3. FOLLOW-UP ENQUIRY [4M]

Q2(b): 4-row tabular enquiry: Detail, Historical Question, Source Type, and Clinical Value (5 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918)	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles <i>Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?</i>	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses <i>How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?</i>	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection <i>Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?</i>	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS <i>How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?</i>	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays <i>How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?</i>	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction <i>How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?</i>	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT–NOV 1914 • 1st Battle of Ypres & The Onset of Static Trench Warfare

Key Topic 5.1

The mobile war ends as both armies dig continuous defensive lines from the English Channel to Switzerland. British troops defend the vulnerable, low-lying Ypres Salient, surrounded on three sides by German artillery on high ground, with an impenetrable clay water table just two feet below the surface.

1915 • Chlorine Gas at 2nd Ypres & The Brodie Steel Helmet

Key Topic 5.2 • 5.3

In April 1915, the German Army launches the first chemical chlorine gas cloud attack at 2nd Ypres, causing fatal pulmonary oedema. Soldiers improvise urine pads before British PH flannel hoods are issued. John Brodie patents the pressed-steel helmet in late 1915, slashing fatal head wounds by 80%.

DEC 1915 • Robert Jones Introduces the Thomas Splint

Key Topic 5.5

Compound femur fractures from artillery shrapnel carry an 80% death rate due to muscle spasm driving bone ends through femoral arteries. Robert Jones trains frontline RAMC officers to fit Hugh Owen Thomas’s splint at the aid post, applying mechanical traction to pull bone ends straight and slashing mortality to 20%.

LIVING TIMELINE • PART 2: SURGICAL INNOVATION & BLOOD STORAGE (1916–1918)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • Battle of the Somme, Debridement & Carrel-Dakin Irrigation

Key Topic 5.1 • 5.5

The British Army suffers 57,470 casualties on Day 1 of the Somme, overwhelming Casualty Clearing Stations. Surgeons abandon Listerian superficial antiseptics for wide wound debridement (excision) to excise dead necrotic tissue before anaerobic gas gangrene develops, coupled with continuous Carrel-Dakin bleach irrigation.

1917 • Arras Underground Hospital & The Mud Logistics of Passchendaele

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In which century BC did Hippocrates practise clinical observation?
2. Name the four bodily fluids in humoural medicine.
3. Which Roman physician developed the Theory of Opposites?
4. In what year did the Black Death reach England?
5. Which Renaissance anatomist published De Fabrica in 1543?
6. Who proved blood circulates through the body in 1628?
7. Which physician classified diseases through bedside observation?
8. In what year was the Royal Society founded?
9. What term describes foul air believed to cause disease?
10. What religious institution monopolized medical copying in the Middle Ages?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Write a concise historical explanation demonstrating the tactical medical difference between a communication trench and a sap.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the system of communication trenches on the Western Front. [2 marks]
Examiner Guidance: Consider how communication trenches connected frontline trenches to dressing stations and supply lines.
Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). [2 marks]
Examiner Guidance: Consider the chalk geology, lighting, water supply, and surgical capacity 20 metres underground.
Feature 2 • Identify another distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE A CONTEMPORARY AERIAL RECONNAISSANCE • ROYAL FLYING CORPS

1917 |



Royal Flying Corps aerial photograph showing the intricate zig-zag traverses of frontline, support, and communication trenches near Ypres, criss-crossed with artillery shell craters.

Provenance: Official military aerial photograph taken by the Royal Flying Corps, British Sector near Ypres, 1917. Imperial War Museum Collection.

Historical Context: An authentic aerial reconnaissance photograph taken by the Royal Flying Corps in 1917 showing the complex defensive network of British frontline, support, and communication trenches in the Ypres sector, displaying the characteristic zig-zag traverse design and dense shell cratering.

Hinge Question: Study Source A. How does the zig-zag design of the trenches shown in the photograph explain how British defensive engineers sought to protect soldiers from artillery shrapnel and machine-gun fire?

Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & UTILITY

Explain how the aerial photograph clearly shows the zig-zag traverse design, communication lines, and support saps.

2. PROVENANCE & MOTIVE

Evaluate the RFC photographic intelligence: highly accurate for cartography, but taken from high altitude under combat conditions.

3. CONTEXT & LIMITATIONS

3. CONTEXT & LIMITATIONS
Cross-reference with knowledge of mud, water tables at Ypres, and the difficulty stretcher bearers faced turning sharp right-angle corners.

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into trench layout because... • Specifically, the aerial perspective reveals... • In terms of provenance, as an official RFC reconnaissance photograph... • However, the source is limited because it cannot show... • Therefore...

KEY VOCABULARY BANK:

Royal Flying Corps • zig-zag traverses • communication trench • Ypres Salient •
Messines Ridge • water table • duckboards • parapet • parados • fire-step

Task: Using the structure strip above, write your analytical source utility evaluation below (continue on Pages 6–7 for full timed assessment):

[illegible]

TIMELINE MISSION

Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table on Milestone 1

← Pages 2-3

EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES •
HISTORIC ENVIRONMENT

6/28 Ypres Salient: surrounded on three sides by German artillery on Messines Ridge; duckboards were life preservers.

KT5.1: The Theatre of War: The British Sector, Trench Geography

**INDEPENDENT PRACTICE • TIMED EXAM
RESPONSE**

[illegible]

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Which French chemist published the Germ Theory of Disease in 1861?
2. Which German scientist identified the specific bacteria for anthrax (1876) and TB (1882)?
3. Who discovered the smallpox vaccination in 1796?
4. Which pioneer transformed sanitation at Scutari Hospital in 1854?
5. Which Scottish doctor discovered the anaesthetic properties of chloroform in 1847?
6. Who pioneered antiseptic carbolic acid spray in surgery in 1865?
7. Who proved cholera was waterborne via the Broad Street pump in 1854?
8. What year was the Second Public Health Act passed in Britain?
9. What was the primary danger of static warfare in the low-lying Ypres Salient?
10. Why were trenches dug in zig-zag traverses rather than straight lines?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how trench foot differed biologically from trench fever, identifying the distinct cause and preventive measure for each.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. [2 marks]

Examiner Guidance: Consider the buddy-system inspections, grease rubbing routines, and spare sock requirements.

Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the causes of trench fever among British troops. [2 marks]

Examiner Guidance: Identify the parasite vector (body lice) and how lice faeces entered scratched skin.

Feature 2 • Identify another distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

ASSESSMENT

SOURCE A CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT

July 1916



British infantrymen of 'A' Company, 11th Battalion, Cheshire Regiment, resting wrapped in waterproof groundsheets on the duckboards and dugouts of a captured trench at Ovillers-la-Boisselle, while a sentry remains on duty on the parapet.

Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke during the Battle of the Somme, July 1916. Imperial War Museum Collection.

Historical Context: An authentic wartime photograph taken in July 1916 during the Battle of the Somme, capturing the extreme physical exhaustion and cramped living conditions of British infantrymen in frontline trenches. Soldiers spent days without sleep or hot food, sheltering in shallow dugouts and on damp duckboards under waterproof groundsheets, creating conditions where body lice (which transmitted trench fever) and extreme physical exhaustion rapidly spread.

Hinge Question: Study Source A. How does this photograph illustrate the harsh living conditions in frontline trenches that made soldiers vulnerable to physical exhaustion and non-combat illnesses like trench fever?

Study Source A. How could you follow up Source A to find out more about the living conditions and non-combat illnesses experienced by British soldiers in frontline trenches on the Western Front?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

trench foot • whale oil • buddy-system • gangrene • trench fever • body lice (Pediculus humanus) • delousing stations • dysentery • chloride of lime • latrine inspection

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives (continue on Pages 10–11 for full timed response):

TIMELINE MISSION Draw soldiers applying whale oil and delousing garments on Milestone 2

← **Pages 2–3**

Non-Combat Illnesses

**EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES •
HISTORIC ENVIRONMENT**

[illegible]

Illnesses

**INDEPENDENT PRACTICE • TIMED EXAM
RESPONSE**

[illegible]

KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What percentage of Western Front wounds were caused by artillery shells and shrapnel?

3. In what year was the British pressed-steel Brodie helmet first issued?

5. In which battle did the German Army first deploy chlorine poison gas in April 1915?

7. Which gas introduced in 1917 caused internal and external blistering and blindness?

9. Who proved that cholera was caused by contaminated water rather than miasma?
2. Which bacterium in heavily manured Flanders soil caused fatal gas gangrene?

4. By what percentage did the Brodie helmet reduce fatal shrapnel head wounds?

6. What emergency measure did British troops use before gas masks were invented?

8. What antiseptic chemical did Joseph Lister pioneer in 1865?

10. What was the daily routine used in British trenches to prevent trench foot?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain the crucial medical difference between wound contamination from shrapnel and chemical tissue destruction from mustard gas.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. [2 marks]

Examiner Guidance: Distinguish between chlorine (suffocation/pulmonary oedema), phosgene, and mustard gas (blistering).

Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the design of the British Brodie steel helmet. [2 marks]

Examiner Guidance: Consider the pressed-steel construction, shallow brim, and internal lining to deflect shrapnel.

Feature 2 • Identify another distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE B CONTEMPORARY PHOTOGRAPHIC EVIDENCE • MACHINE GUN CORPS**July 1916**

A British Machine Gun Corps crew operating a Vickers machine gun from an improvised trench position near Ovillers on the Somme while wearing issue Phenate-Hexamine (PH) anti-gas helmets with exhale valves.

Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke near Ovillers during the Battle of the Somme, July 1916. Imperial War Museum Collection.

Historical Context: An authentic wartime photograph taken in July 1916 during the Battle of the Somme showing British machine gunners in combat wearing the issue PH anti-gas helmet. Made of double-layered flannel impregnated with sodium phenate and hexamine to chemically neutralize chlorine and deadly phosgene gas, the helmet was tucked into the tunic collar. While it prevented fatal lung damage, the mica eyepieces easily misted over and breathing through the tube during active combat was exhausting.

Hinge Question: Study Source B. How does this photograph illustrate both the vital protective value of early anti-gas helmets and the severe physical difficulties experienced by soldiers fighting while wearing them?

Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & ARTIFACT UTILITY

Explain how the PH hood protected against chlorine and phosgene, featuring glass eyepieces and an exhalation valve.

2. PROVENANCE & MOTIVE

Evaluate the artifact: official War Office standard issue, demonstrating rapid British industrial response to chemical warfare.

3. CONTEXT & COMBAT LIMITS

Cross-reference with knowledge: hot, suffocating, prone to fogging, and completely ineffective against mustard gas introduced in 1917.

ANALYTICAL CONNECTIVES:

Source B is useful for investigating gas defence because... • Specifically, the physical helmet proves that... • In terms of provenance, as a standard-issue War Office artifact... • However, Source B is limited because it cannot show... • Consequently...

KEY VOCABULARY BANK:

high-explosive artillery • shrapnel balls • gas gangrene • *Clostridium welchii* • Brodie steel helmet • chlorine gas (1915) • phosgene • mustard gas (1917) • PH helmet • Small Box Respirator (1916)

Task: Using the structure strip above, write your analytical source utility evaluation below (continue on Pages 14–15 for full timed assessment):

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

TIMELINE MISSION Sketch the Brodie steel helmet and British PH gas hood on Milestone 2

← Pages 2-3

KT5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection

**EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES •
HISTORIC ENVIRONMENT**

[illegible]

Infection

**INDEPENDENT PRACTICE • TIMED EXAM
RESPONSE**

[illegible]

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. How many stretcher bearers were assigned to an infantry battalion of 1,000 men?

2. How many bearers were typically required to carry one wounded man in Passchendaele mud?

3. How far behind the frontline was a Regimental Aid Post (RAP) usually located?

4. Could major surgery be performed at a Regimental Aid Post?

5. Which medical unit operated Advanced and Main Dressing Stations (ADS/MDS)?

6. What station in the evacuation chain was the primary surgical center near rail lines?

7. What three triage categories were used at Casualty Clearing Stations?

8. What female voluntary organisation drove motor ambulances on the Western Front?

9. Why were canal barges preferred for transporting chest and brain wound patients?

10. Where were large British Base Hospitals located in France?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain why the triage classification system at Casualty Clearing Stations was necessary when dealing with mass-casualty offensives.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. [2 marks]

Examiner Guidance: Explain how casualties were sorted into Walking Wounded, Immediate Surgery, and Moribund to maximize survival.

Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. [2 marks]

Examiner Guidance: Consider ambulance driving, mobile soup kitchens, and transporting wounded through artillery fire.

Feature 2 • Identify another distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

SOURCE A
CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT

August 1917



Provenance: Official British Army photograph taken by Lieutenant John Warwick Brooke near Boesinghe during the Third Battle of Ypres, 1 August 1917. Imperial War Museum Collection.

Historical Context: Official photograph taken by British war photographer Lieutenant John Warwick Brooke on 1 August 1917 during the Third Battle of Ypres (Passchendaele), capturing the grueling physical ordeal experienced by stretcher bearers navigating liquid-mud battlefields under fire, requiring six men rather than four to haul a single stretcher.

Hinge Question: *Study Source A. How does this photograph illustrate why transporting wounded soldiers across devastated battlefields to dressing stations required up to six bearers and took many exhausting hours?*

A team of six British stretcher bearers struggling through deep, liquid Flanders mud to carry a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:
A key detail to follow up is... • *The question I would ask is...* • *The specific historical source type needed is...* • *This source would help answer my question because...*

KEY VOCABULARY BANK:
 RAMC • FANY • Regimental Aid Post • Advanced Dressing Station • Casualty Clearing Station • Base Hospital • triage system • motor ambulance • hospital train • canal barge

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives (continue on Pages 18–19 for full timed response):

TIMELINE MISSION
Draw the 6 sequential stages of the Chain of Evacuation on Milestone 5

← Pages 2–3

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES • HISTORIC ENVIRONMENT

[illegible]

KT5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS

INDEPENDENT PRACTICE • TIMED EXAM RESPONSE

[illegible]

KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What was the mortality rate for compound femur fractures in 1914 before the Thomas Splint?

3. To what percentage did the Thomas Splint reduce compound femur mortality by 1916?

5. What surgical term describes cutting away dead, infected tissue from a gunshot wound?

7. Why did Carrel-Dakin solution need to be made fresh in laboratories every 6 hours?

9. Name one limitation of mobile X-ray units used at Casualty Clearing Stations.
2. Who introduced the Thomas Splint to frontline medical officers in December 1915?

4. How did the Thomas Splint prevent fatal internal haemorrhage in fractured thighs?

6. What antiseptic irrigation solution was used continuously through perforated rubber tubes?

8. Who discovered X-rays in Germany in 1895?

10. What was the role of Marie Curie in mobile radiology during the Great War?

KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how wound debridement combined with Carrel-Dakin irrigation prevented fatal gas gangrene in high-explosive wounds.

EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the Thomas Splint used on the Western Front. [2 marks]

Examiner Guidance: Explain the padded metal ring fitted against the groin and the windlass traction system on the foot.

Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. [2 marks]

Examiner Guidance: Explain the system of perforated rubber tubes irrigating the deep wound with sodium hypochlorite bleach.

Feature 2 • Identify another distinct historical feature [1 mark]:

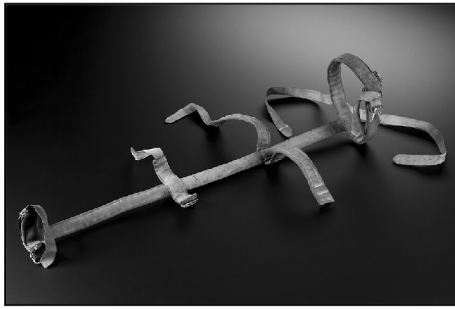
Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY ASSESSMENT

SOURCE A CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE

c.1916



An authentic surviving British Army Thomas Splint, showing the padded oval ring fitted against the groin, rigid metal side bars, and canvas support straps used to maintain traction and immobilize broken thigh bones.

Provenance: Original First World War Thomas Splint preserved in the Wellcome Historical Medical Collection, London. Designed by Hugh Owen Thomas and introduced to the Western Front in 1915–1916 by Robert Jones.

Historical Context: An authentic surviving artifact of the Thomas Splint championed on the Western Front by Robert Jones from late 1915, which revolutionized battlefield trauma care. Before its widespread introduction, 80% of soldiers suffering compound fractures of the femur died from blood loss, shock, and internal hemorrhage caused by broken bone ends grinding during transit. The rigid splint held the leg taut under traction, stabilizing the limb throughout evacuation and reducing femur mortality to under 20% by 1917.

Hinge Question: Study Source A. How does the mechanical design of the Thomas Splint shown in the photograph explain why survival rates for fractured femurs rose from 20% in 1914 to 80% by 1917?

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

STRUCTURE STRIP • 3-STEP UTILITY ANALYSIS

1. CONTENT & UTILITY

Explain how Source A illustrates the Thomas Splint: groin ring, steel framework, and traction cord pulling the leg taut to prevent bone grinding.

2. PROVENANCE & TONE

Evaluate the photograph: produced for medical training manuals, demonstrating correct surgical application under clean, controlled conditions.

3. CONTEXT & LIMITS

Cross-reference with knowledge: Robert Jones 1915, 80% down to 20% mortality; note limitations—
 photograph cannot convey agony or muddy dugout application.

ANALYTICAL CONNECTIVES:

Source A is useful for an enquiry into limb treatments because... • Specifically, the photograph demonstrates... • In terms of provenance, as an RAMC instructional image... • However, the source is limited because... • Therefore...

KEY VOCABULARY BANK:

Thomas Splint • Hugh Owen Thomas • Robert Jones • compound fracture • femoral artery • mechanical traction • wound debridement • Carrel-Dakin solution • delayed primary closure • mobile X-ray unit

Task: Using the structure strip above, write your analytical source utility evaluation below (continue on Pages 22–23 for full timed assessment):

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

TIMELINE MISSION Sketch the Thomas Splint groin ring and traction cord on Milestone 3

← Pages 2-3

KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES • HISTORIC ENVIRONMENT

**EXTENDED SOURCE EVALUATION • DISCIPLINARY
NOTES • HISTORIC ENVIRONMENT**

[illegible]

KT5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays

INDEPENDENT PRACTICE • TIMED EXAM RESPONSE

[illegible]

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Who discovered the main ABO blood groups in Vienna in 1901?

2. What chemical was discovered by Albert Hustin in 1914 to prevent blood clotting?

3. What substance did Rous and Turner add in 1916 to preserve blood for up to 4 weeks?

4. Who established the world's first stored blood depot at the Battle of Cambrai in 1917?

5. What blood group was chosen for universal transfusion in Robertson's blood depot?

6. Which American surgeon pioneered precise neurosurgery using local anaesthetic on the Western Front?

7. To what percentage did Harvey Cushing reduce brain surgery mortality in 1917?

8. Which New Zealand-born surgeon pioneered facial reconstructive plastic surgery at Sidcup?

9. What surgical skin-transfer technique did Harold Gillies invent in 1917?

10. What hospital in Kent became the world centre for British plastic reconstruction?

• KEY VOCABULARY APPLICATION TASK

TERMINOLOGY

Explain how sodium citrate combined with refrigeration revolutionized blood transfusion between 1914 and 1917.

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS]

[2 × 2 MARKS]

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. To score 2/2, the supporting detail must directly explain and develop the named feature.

1(a) Describe one feature of the blood depot established by Oswald Robertson at the Battle of Cambrai in 1917. [2 marks]
Examiner Guidance: Consider the use of sodium citrate, ice chests, Type O blood, and treating shock before surgery.
Feature 1 • Identify a distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. [2 marks]
Examiner Guidance: Explain the tubed pedicle method of maintaining blood supply while grafting facial tissue.
Feature 2 • Identify another distinct historical feature [1 mark]:

Supporting Detail • Develop with precise factual explanation [1 mark]:

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 PRIMARY ARCHIVAL SOURCE • SECTION A ENQUIRY mins]

SOURCE A
PRIMARY MEDICAL ARTIFACT • CAMBRAI BLOOD DEPOT
1917

The authentic wooden ice chest, glass collection flasks, rubber delivery tubing, and sodium citrate solutions utilized by Captain Oswald Robertson to establish the world's first blood bank during the Battle of Cambrai in November 1917.

Provenance: Original portable blood transfusion outfit preserved in the Royal Army Medical Corps Historical Collection, London. Utilized during the Battle of Cambrai, November 1917.

Historical Context: An authentic surviving artifact photograph of Captain Oswald Robertson's blood storage and transfusion kit used at the Battle of Cambrai in November 1917. Robertson collected blood in sterile glass bottles containing sodium citrate and dextrose (glucose) and stored them inside an insulated wooden ice chest, enabling stored blood to be transfused up to 26 days after collection.

Hinge Question: Study Source A. How does this blood storage apparatus explain how medical officers at the Battle of Cambrai were able to treat soldiers suffering from severe shock before operations began?

Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917?

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would look for:	
How this might help answer my question:	

SENTENCE STEMS:
A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:
Karl Landsteiner • ABO blood groups • sodium citrate (Hustin 1914) • glucose preservative (Rous & Turner 1916) • Oswald Robertson • Battle of Cambrai (1917) • blood depot • Harvey Cushing • Harold Gillies • tubed pedicle graft

Enquiry Rationale: Explain in full sentences why your chosen source type is historically reliable and superior to alternatives (continue on Pages 26–27 for full timed response):

TIMELINE MISSION
Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle on Milestone 6

← Pages 2–3

KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction

EXTENDED SOURCE EVALUATION • DISCIPLINARY NOTES
• HISTORIC ENVIRONMENT

[illegible]

KT5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction INDEPENDENT PRACTICE • TIMED EXAM RESPONSE

[illegible]

STUDENT ASSESSMENT RECORD & PROGRESS TRACKER

PUPIL NAME: _____ CLASS: _____ TARGET GRADE: _____

Date Completed	Lesson • Specification Focus	Q1 Feature Drills [4m]	Q2 Source Practice	Lesson Total
	KT5.1: Trench System & Battles	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.2: Trench Illnesses & Mud	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
	KT5.3: Combat Trauma & Gas	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.4: Evacuation Chain & CCS	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
	KT5.5: Splints, Debridement, X-Rays	Q1: [/ 4]	Q2a Util: [/ 8]	[/ 12]
	KT5.6: Blood Storage & Reconstruction	Q1: [/ 4]	Q2b Follow: [/ 4]	[/ 8]
SECTION A ASSESSMENT TOTALS		Q1 Total: [/ 24]	Q2 Total: [/ 36]	[/ 60]

TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [1 • 2 • 3 • 4 • 5]

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

Teacher Signature: _____Date: _____

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT5.1

Trenches & Mud

SCAN TO QUIZ

Score: [/ 8]

KT5.2

Trench Illness

SCAN TO QUIZ

Score: [/ 8]

KT5.3

Trauma & Gas

SCAN TO QUIZ

Score: [/ 8]

KT5.4

Evacuation

SCAN TO QUIZ

Score: [/ 8]

KT5.5

Splints & X-Rays

SCAN TO QUIZ

Score: [/ 8]

KT5.6

Blood & Plastic

SCAN TO QUIZ

Score: [/ 8]